# Automated Retail Analytics

## Sample Report — Retail Sales & Customer Analysis

Data Source

E-commerce Sales & Customer Data (2009–2011)

Analyses Included

① Data Cleaning & Quality Assurance

② Sales Trend Analysis (Monthly / Weekly / YoY)

③ Product Performance & Pareto Analysis

④ Customer Segmentation (RFM)

Report Date: October 12, 2026

Author: Senior Data Scientist

Available for freelance data analysis engagements

This is a sample report. Fully customised to your business data and goals.

## Background & Objective

An e-commerce retailer held two years of sales and customer data but lacked the tools to turn it into decisions. This report structures the data across three lenses — revenue trends, product performance, and customer behaviour — and surfaces next actions.

| £17.7M | 5,878 | Top 20% |
|---|---|---|
| Total Revenue | Customers Analysed | of products drive 80% |
| 2009–2011 | After data cleaning | of revenue (Pareto) |

## Key Findings (3)

1. Revenue shows strong November seasonality, consistent across both years.
   → Stock and staffing must be prepared by mid-September at the latest.

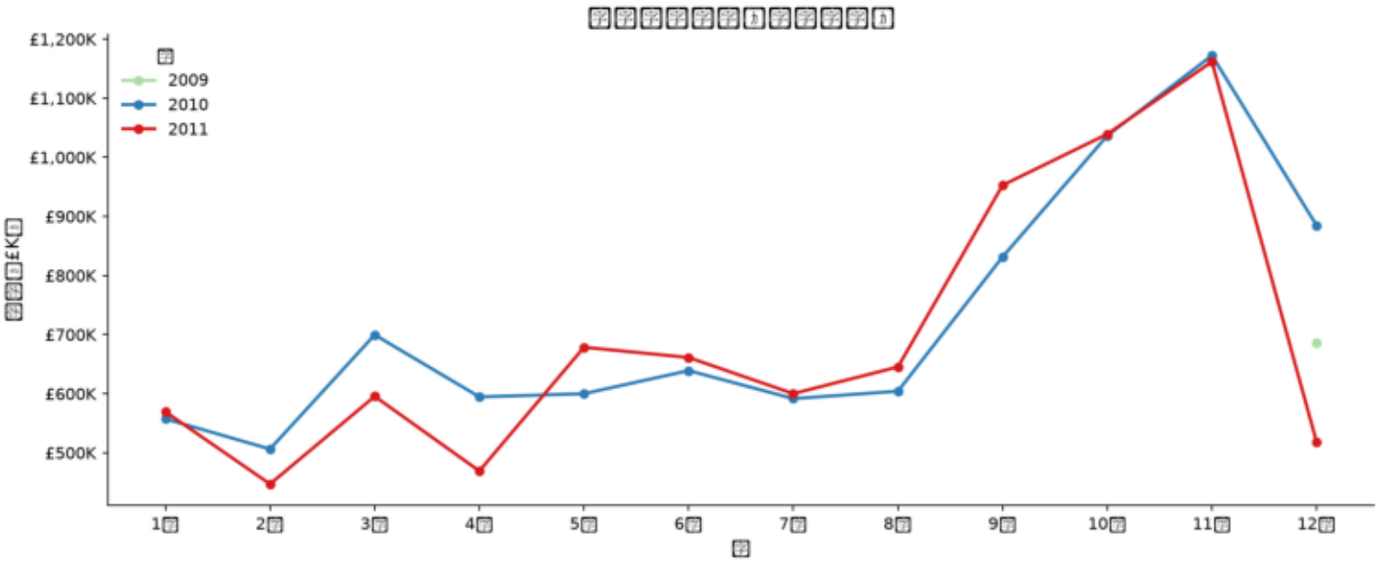
2. The top 20% of products generate the majority of revenue (Pareto principle).
   → Preventing stockouts on hero SKUs is the highest-leverage inventory action.

3. Champion customers are a small share of the base but contribute a disproportionate share of revenue. Retaining them delivers the best return on any retention spend.

## Dataset Snapshot

| | |
|---|---|
| Analysis Period | December 2009 – December 2011 |
| Transaction rows | ˜805,000 (after cleaning) |
| Unique customers | 5,878 |
| Unique products | 5,283 SKUs |
| Markets | UK-centric · 40+ countries |

## Monthly Revenue — Year-on-Year Comparison

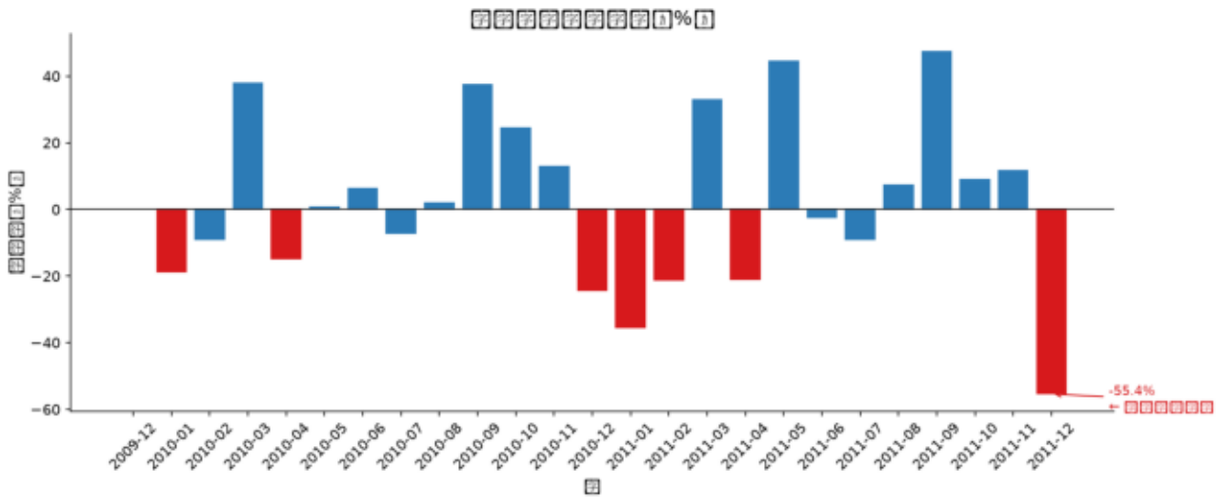



## Analysis Commentary

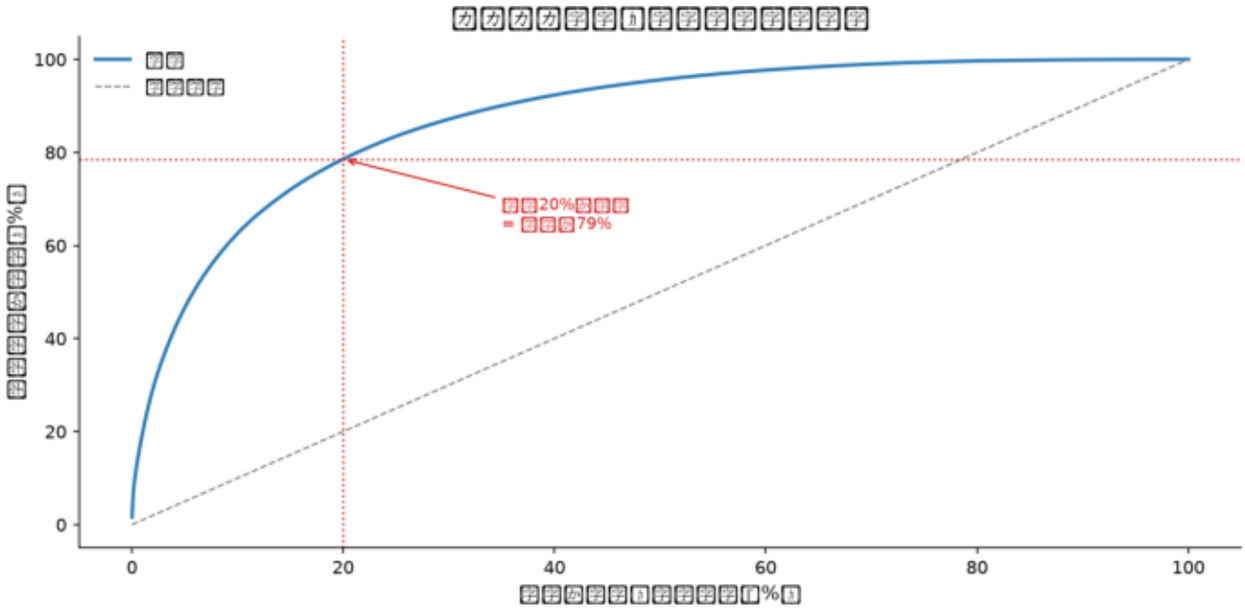
- A November revenue spike is confirmed for two consecutive years, driven by holiday-season demand.
- Overall revenue grew year-on-year in 2011, confirming the underlying business growth trajectory.

→ Inventory and staffing prep must begin by mid-September at the latest. Acting early on holiday demand is the single biggest revenue lever.

## Month-over-Month Revenue Change (%)



## Pareto Analysis — Revenue Concentration

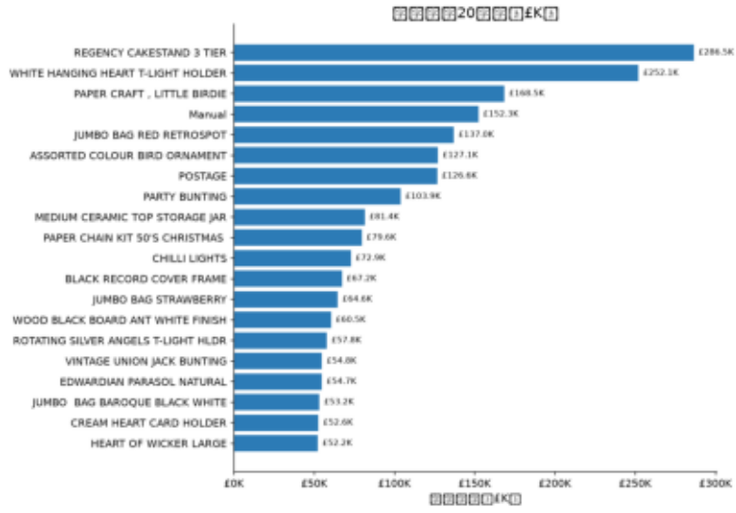



## Analysis Commentary

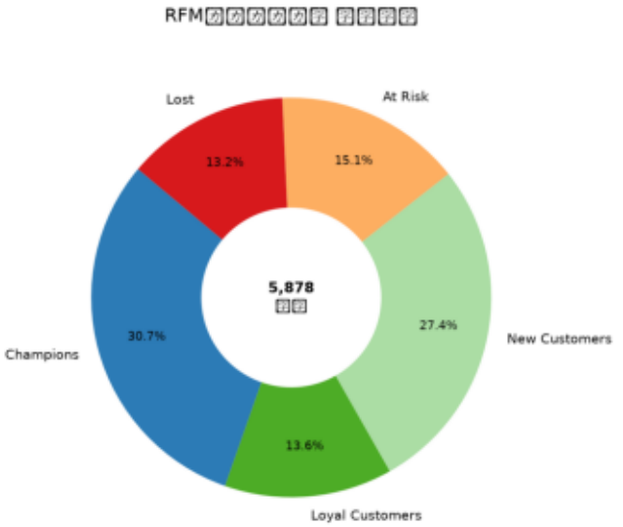
- The Pareto principle holds: the top 20% of SKUs account for the large majority of revenue.
- High-order-frequency products with low revenue share are candidates for a modest price increase.

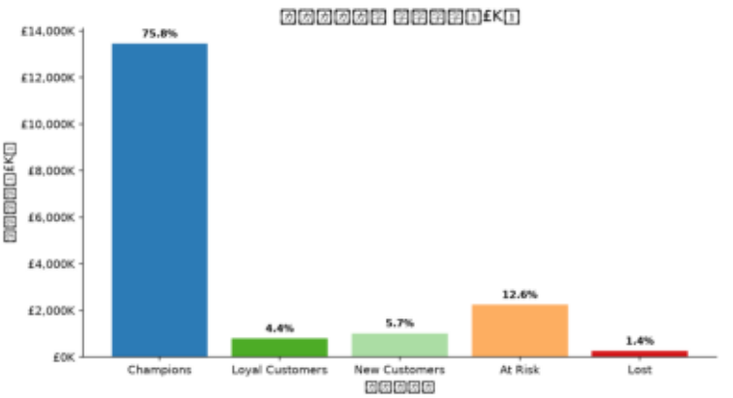
→ Zero stockouts on the top 10 SKUs is the single most impactful supply-chain target — these items alone represent roughly 40% of total revenue.

## Top 20 Products by Revenue



## Segment Distribution — Customer Count & Revenue Share




## Recommended Actions by Segment

| | | |
|---|---|---|
| Champions | Recent · Frequent · High-value | VIP perks, early access to new products |
| Loyal Customers | Repeat buyers, high engagement | Loyalty programme, subscription discounts |
| At Risk | Past high-value, recently quiet | Limited-time coupon within 30 days |
| New Customers | First or second purchase | 7-day post-purchase follow-up email |
| Lost | Long-lapsed, no recent activity | Win-back campaign or retire from list |

→ Champions are ~31% of customers but drive ~76% of revenue. Protecting this group has a higher ROI than acquiring new customers.

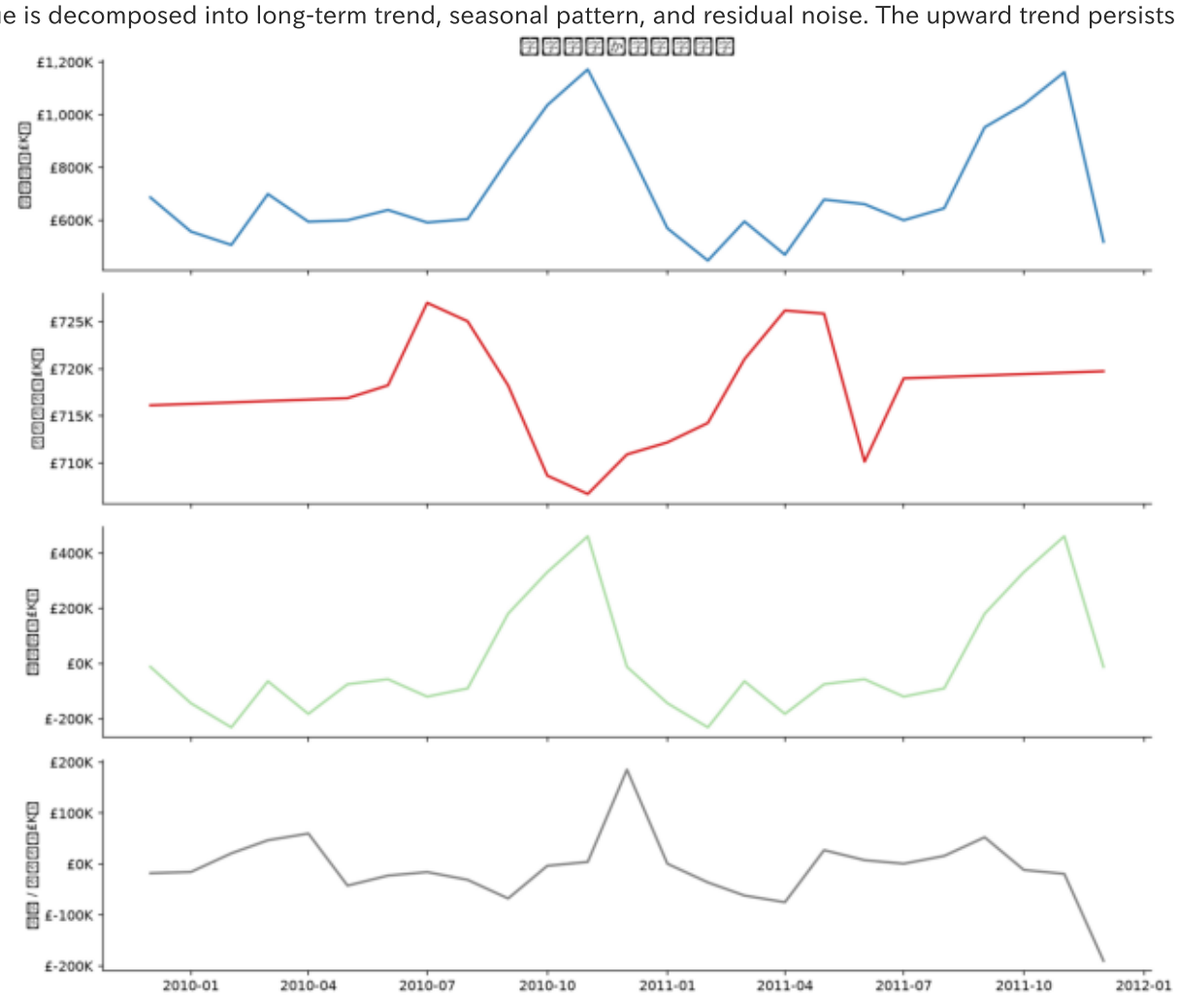
## RFM Score Heatmap by Segment




Average R (Recency), F (Frequency), and M (Monetary) scores per segment. Champions score high across all three dimensions; Lost customers score low across all three.

## Seasonal Decomposition (Trend · Seasonality · Residual)

Revenue is decomposed into long-term trend, seasonal pattern, and residual noise. The upward trend persists after removing seasonality — confirming genuine underlying growth.



## Methodology

This report was produced using industry-standard analytical methods. Results are presented in plain language — no technical background required.

### Data Cleaning

Removed cancellations, missing customer IDs, and invalid prices. Only reliable records enter the analysis.

### Trend Analysis

Monthly and weekly aggregation with year-on-year comparison. Seasonal decomposition isolates structural growth.

### Pareto Analysis

Visualises revenue concentration. Identifies which products and customers deserve the most focus.

### RFM Segmentation

Scores customers on Recency, Frequency, and Monetary value, then classifies them into 5 actionable groups.

## Enquiries & Engagements

Active data scientist. Available for freelance analysis projects.

- Send your Excel or CSV — that's all we need to get started
- Delivered as a branded PDF or Excel workbook
- Plain-language commentary with every chart
- Happy to scope open-ended or unclear briefs

Get in touch — first conversation is free.

This is a sample report. All analyses are customised to your actual business data and objectives.